cheeps when they are lost or cold. This usually has the immediate effect of summoning the mother, who leads the chick back to the main clutch. This behaviour could be said to have evolved for mutual benefit, in the sense that natural selection has favoured babies that cheep when they are lost, and also mothers that respond appropriately to the cheeping.

If we wish to (it is not really necessary), we can regard signals such as the cheep call as having a meaning, or as carrying information: in this case 'I am lost.' The alarm call given by small birds, which I mentioned in Chapter 1, could be said to convey the information 'There is a hawk.' Animals who receive this information and act on it are benefited. Therefore the information can be said to be true. But do animals ever communicate false information; do they ever tell lies?

The notion of an animal telling a lie is open to misunderstanding, so I must try to forestall this. I remember attending a lecture given by Beatrice and Allen Gardner about their famous 'talking' chimpanzee Washoe (she uses American Sign Language, and her achievement is of great potential interest to students of language). There were some philosophers in the audience, and in the discussion after the lecture they were much exercised by the question of whether Washoe could tell a lie. I suspected that the Gardners thought there were more interesting things to talk about, and I agreed with them. In this book I am using words like 'deceive' and 'lie' in a much more straight-forward sense than those philosophers. They were interested in conscious intention to deceive. I am talking simply about having an effect functionally equivalent to deception. If a bird used the 'There is a hawk' signal when there was no hawk, thereby frightening his colleagues away, leaving him to eat all their food, we might say he had told a lie. We would not mean he had deliberately intended consciously to deceive. All that is implied is that the liar gained food at the other birds' expense, and the reason the other birds flew away was that they reacted to the liar's cry in a way appropriate to the presence of a hawk.

Many edible insects, like the butterflies of the previous chapter, derive protection by mimicking the external appearance of other distasteful or stinging insects. We ourselves are often fooled into thinking that yellow and black striped hover-flies are wasps. Some bee-mimicking flies are even more perfect in their deception. Predators too tell lies. Angler fish wait patiently on the bottom of the sea, blending in with the background. The only conspicuous part is a wriggling worm-